Gyroid Cube with 3 × 3 × 3 Openings

Design modification, validation and PLA 3D-printing report

ME3305 Manufacturing Technology II · 29 October 2026

Item	Result
Final STL	stl/gyroid_3x3_40mm.stl (400,000 triangles, 20.0 MB)
External size (measured)	40.000 × 40.000 × 40.000 mm
Openings along X / Y / Z (measured on STL)	9 (3×3) / 9 (3×3) / 9 (3×3) see-through openings; minimum clear Ø 4.84 mm
Openings after slicing (PrusaSlicer 2.9.6 G-code)	9 / 9 / 9 per axis, minimum clear Ø 4.82 mm
Original print file (for comparison)	X/Y: 0 openings (closed side walls); Z: 90 openings of Ø 1.0 mm
Mesh	watertight, 1 body, 0 open/non-manifold edges, 0 self-intersections, 0 degenerate faces
Supports	none needed: sliced with supports OFF, 0 mm support; PrusaSlicer estimate 1h 57m 47s, 14.78 g PLA
Not performed	physical print, strength test, slicer GUI preview on your printer profile

All measured values in this report are read from the generated files at report-build time. **Verified** = measured on the STL or on the PrusaSlicer G-code. **Estimate** = derived from a model or assumption. **Not performed** = stated explicitly; no print, mechanical test or simulation has been run.

1. Original design assessment

The previously printed model came from the revision-2 generator (*src/gyroid.py*, *src/build_final.py*): sheet gyroid, period **a = 8 mm** in a 40 mm cube (**5 periods per axis**), nominal wall 0.9 mm (measured 0.87 mm median), and in the recommended print file 1.25 mm side walls on the four X/Y faces.

Problem reported	Cause found (measured)
Openings too small / hard to see	Viewed along an axis the sheet gyroid has 2 openings per period, so a = 8 mm gives a 10 × 10 grid: 90 whole see-through openings per face plus edge-cut ones, each only Ø 1.0 mm (5.9 % of the face open). The print file's side walls closed the X and Y faces completely: 0 openings along X and Y.
Supports trapped inside	Re-slicing the original file with supports 'everywhere' (PrusaSlicer 2.9.6) generates 201 m of support, of which 17,233 mm³ stands on the lattice's own surfaces in 60 separate pockets – none of it reachable from outside. Its overhangs reach at most 0.58 mm per layer, so supports were not needed in the first place. Estimate with supports: 10h 36m 47s, 52.98 g.

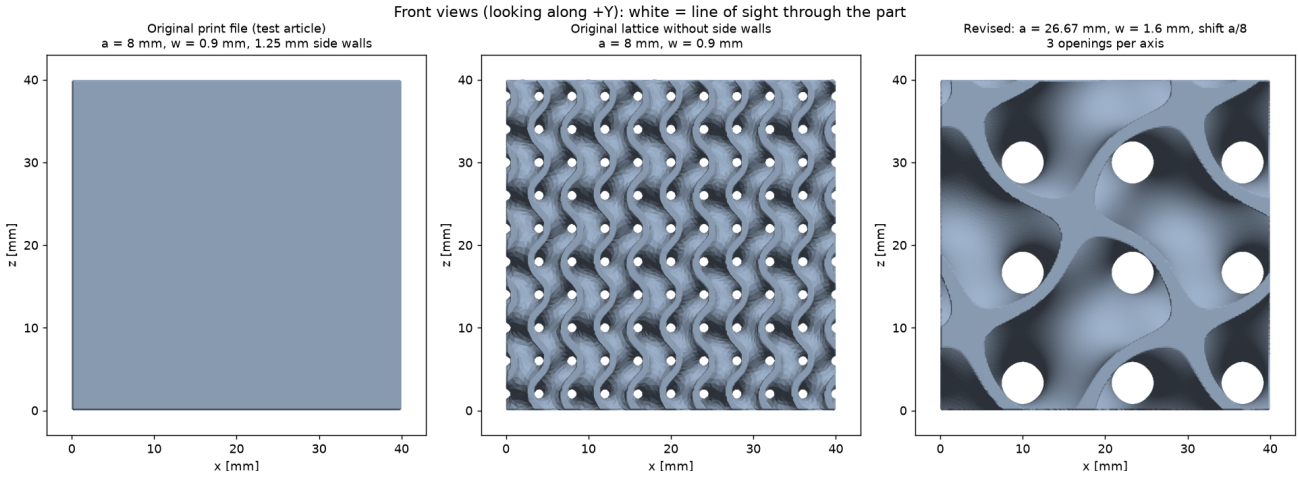


Figure 1 – Front views rendered from the STL files at the same scale (white = straight line of sight through the part). Left: original print file (closed side walls). Centre: original lattice without side walls. Right: revised design.

2. Updated geometry

2.1 Generation method

The existing generator was modified, not replaced. The sheet gyroid keeps its equation and its constant-thickness wall; only the period, the phase (where the cube faces cut the lattice) and the wall thickness change:

g(x,y,z) = sin X·cos Y + sin Y·cos Z + sin Z·cos X, X = 2π(x + s)/a (same for Y, Z); solid where |g| / |grad g| ≤ w/2 (sheet of thickness w around g = 0), intersected with the 40 mm cube. Meshed with marching cubes at 0.2 mm pitch, quadric-decimated with acceptance gates (§4).

2.2 How the unit cell maps to visible openings

A 3 × 3 pattern does not follow from setting 3 unit cells per axis. Projecting the sheet gyroid along a cube axis (rays through the whole part) shows **4 see-through openings per period², on a square grid with spacing a/2**. The number of openings across a face of size L is therefore 2L/a, not L/a:

Configuration	Period a	Periods / axis	Openings per face (whole + edge-cut)	Min clear Ø
original (printed) a=8, w=0.9, shift 0	8.00 mm	5.00	90 + 20	1.00 mm
naive 3 periods a=13.33, w=1.6, shift a/8	13.33 mm	3.00	36 + 0	1.70 mm
3 openings a=26.67, w=1.6, shift a/8 – **selected**	26.67 mm	1.50	9 + 0	4.84 mm
3 openings a=26.67, w=1.6, shift 0 (wrong phase)	26.67 mm	1.50	6 + 6	4.88 mm

Hence 3 openings per axis requires a = 2L/3 = **26.67 mm** (1.5 periods across 40 mm). The phase then decides whether the openings are whole or cut by the cube edges: only a shift of **s = a/8 = 3.33 mm** on all three axes gives 9 whole openings and none cut on every face simultaneously (shift 0 gives 6 whole + 6 cut). The two families of openings seen in perpendicular views are offset by a/4, so they cannot both be centred on a face; each 3 × 3 grid sits 3.3 mm off the face centre, with a 0.9 mm solid border to the nearest cube edge. The external size stays 40 × 40 × 40 mm; no dimensional change was needed.

Final geometry	Value	Status
External dimensions X × Y × Z	40.000 × 40.000 × 40.000 mm	Verified (STL)
Gyroid period a / periods per axis	26.667 mm / 1.50	Design input
Opening cells per axis (opening pitch a/2)	3 × 3 × 3 (pitch 13.33 mm)	Verified (STL)
Phase shift s (all axes)	3.333 mm = a/8	Design input
See-through openings X / Y / Z	9 whole (3×3), 4 edge slivers / 9 whole (3×3), 4 edge slivers / 9 whole (3×3), 4 edge slivers	Verified (STL)
Clear opening diameter (min / max)	4.84 / 4.94 mm	Verified (STL, 0.1 mm voxels)
Open fraction of each face (line of sight)	11.1 %	Verified (STL)
Largest sphere passing face-to-face	9.9 / 9.9 / 9.9 mm (X / Y / Z)	Verified (STL)
Wall thickness, nominal / measured 3D (p1, median, max)	1.6 mm / 1.56, 1.61, 1.70 mm	Verified (STL)
Solid volume / solid fraction	11,627 mm³ / 18.2 %	Verified (STL)
Surface area (internal, both sides)	14,669 mm²	Verified (STL)
PLA mass	14.4 g (at 1.24 g/cm³); slicer: 14.78 g incl. brim	Estimate

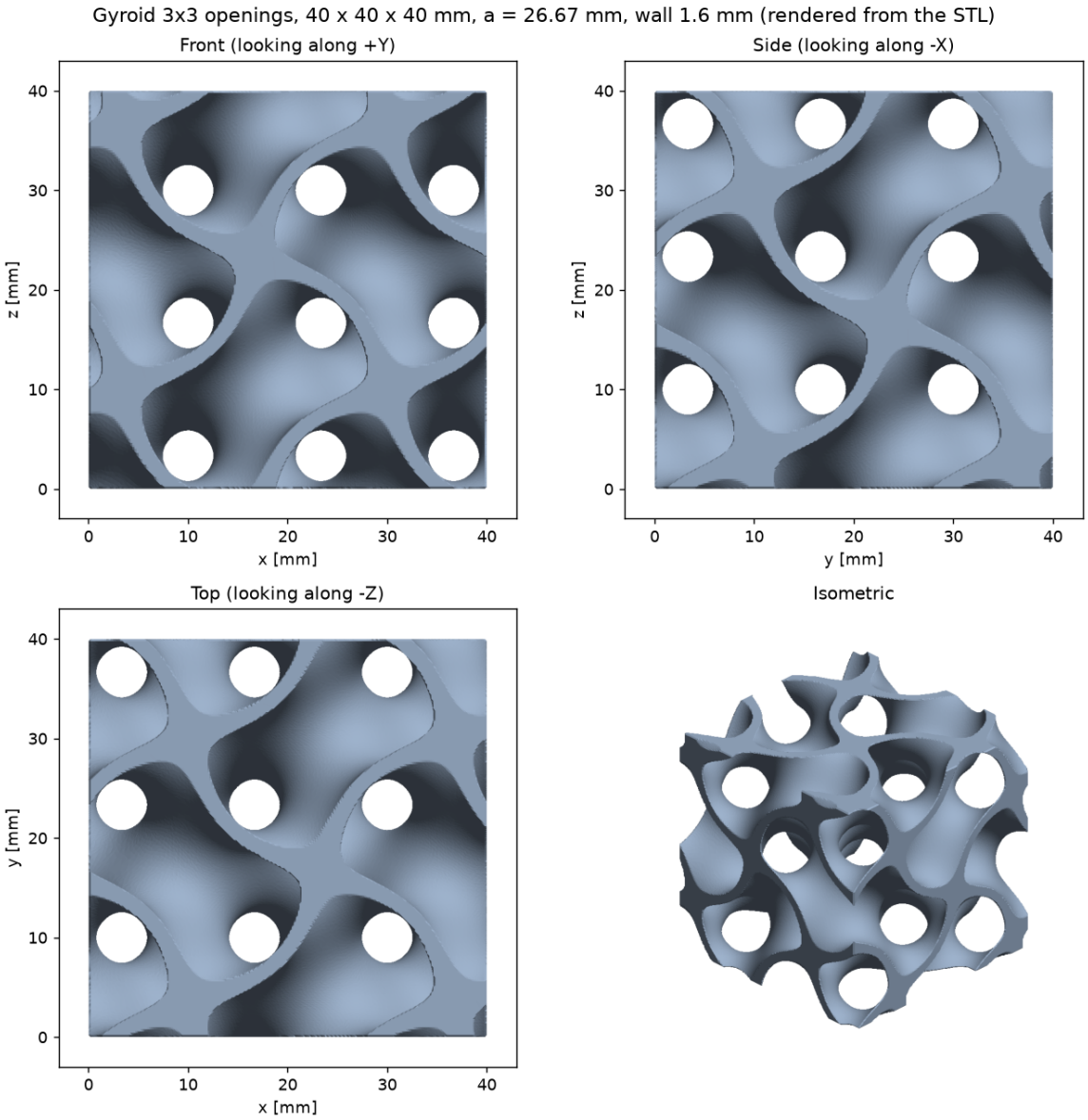


Figure 2 – Front, side, top and isometric views rendered from gyroid_3x3_40mm.stl (orthographic, z-buffered over the real triangles; white = background seen through the part).

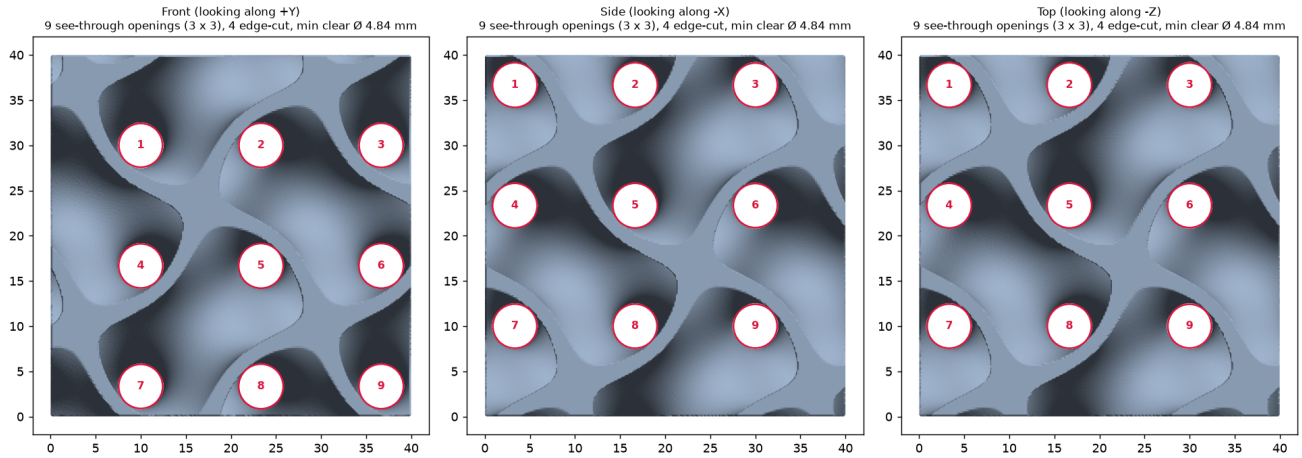


Figure 3 – Every see-through opening detected on the STL, numbered; circles = measured clear diameter.

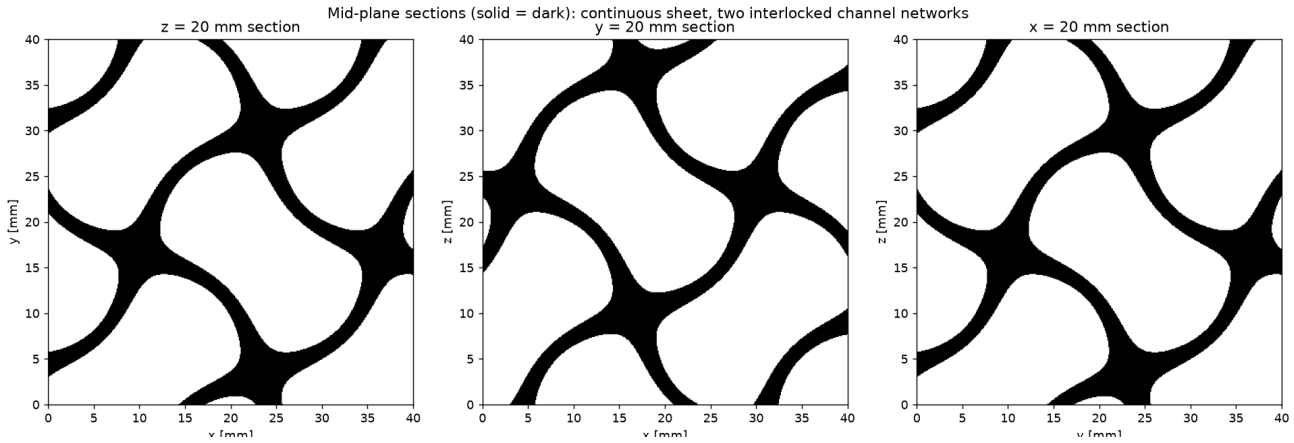


Figure 4 – Mid-plane sections: one continuous sheet separating two interlocked channel networks.

3. Design changes

Parameter	Original (printed)	Updated	Reason
Overall size	40 × 40 × 40 mm	40 × 40 × 40 mm	Unchanged
Gyroid period a	8 mm	26.67 mm	2 openings per period → 3 per axis needs a = 2L/3
Periods per axis	5	1.5	As above
Phase shift	0	a/8 = 3.33 mm	Only phase with 9 whole, uncut openings on all faces
See-through openings per face	X/Y: 0 (side walls), Z: 90	9 on every face (3 × 3)	Primary requirement
Clear opening Ø	1.0 mm	4.84 mm	≈4.8× larger
Side walls (skin)	1.25 mm on X/Y faces	none	They blocked the X and Y openings
Wall nominal / measured median	0.9 / 0.87 mm	1.6 / 1.61 mm	4 lines at 0.4 mm; panels are 3.3× larger, so thicker for stiffness
Channel bottleneck (sphere)	2.58 mm	10.0 mm	Supports and debris can leave; inspection easy
Solid fraction / mass	40.7 % / 32.3 g	18.2 % / 14.4 g	Larger cells, no side walls
Recommended supports	(printed with supports)	OFF	Self-supporting; see §7
STL triangles / size	600,000 / 30 MB	400,000 / 20 MB	Fewer, larger surfaces need fewer triangles

3.1 Wall thickness choice

Nominal wall	Measured p1 / median	Min clear Ø	Lines @ 0.4 mm	Mass (est.)	Assessment
0.9 mm	0.80 / 0.92 mm	5.52 mm	2.2	8.2 g	thin 13 mm panels, flexible
1.2 mm	1.13 / 1.22 mm	5.23 mm	3.0	10.9 g	acceptable
1.6 mm	1.55 / 1.60 mm	4.84 mm	4.0	14.5 g	**selected**: robust panels
2.0 mm	1.87 / 2.01 mm	4.49 mm	5.0	17.9 g	smaller openings, heavier

The panels between openings are about 3.3× wider than before. For a flat plate, keeping the same bending stiffness at 3.3× the span would need about 2.2× the thickness ($t^3 \propto span^2$); the doubly curved gyroid shell is stiffer than a flat plate, so 1.6 mm (1.9× the original 0.85 mm) was chosen as a compromise that still keeps the openings within 8 % of the 1.2 mm option. This is an **engineering estimate**, not a tested strength value.

4. 3D-printability assessment

4.1 Mesh validation (STL reloaded from disk)

Check	Result	Status
Watertight / consistent winding / valid volume	True / True / True	Verified
Connected bodies	1 (genus 34)	Verified
Open edges / non-manifold edges	0 / 0	Verified
Self-intersections (edge–triangle test)	0	Verified
Zero-area faces	0	Verified
Decimation change vs. full mesh	volume -0.002 %, area +0.015 %	Verified
PrusaSlicer import	manifold = yes, 1 part (prusa-slicer --info)	Verified

4.2 Minimum feature sizes

Feature	Measured	Comment
3D wall thickness (>1 mm from faces)	p1 1.56 mm, median 1.61 mm	≥ 3 extrusion lines everywhere
In-plane wall width per 0.2 mm layer	p1 1.30 mm, median 1.80 mm; 0.00 % below 0.8 mm	Prints as 3–4 lines
Strips on the six cube faces	p1 0.40 mm, median 1.72 mm; 0.39 % below 0.4 mm	Thin only where a wall meets a face at a grazing angle
Knife-edge slivers on the vertical cube edges	3 specks, each ≤ 0.06 mm²	Smaller than one line; slicer drops or blobs them; cosmetic
Clear openings after slicing	min Ø 4.82 mm (all 27 present)	Verified on G-code

4.3 Overhangs and bridging

Gyroid sheets have no horizontal down-facing areas except near saddle points. Measured per layer: 11.5 % of the surface faces down steeper than 45° and 1.8 % steeper than 70°. The largest region printed more than one line width beyond the layer below is 3.1 mm² with a reach of 1.24 mm (saddle ceilings at z ≈ 5.9, 25.9 mm); the original reached 0.58 mm. These are short cantilevers, far below practical PLA bridging limits (~10 mm), and PrusaSlicer handled them with 771 mm of overhang perimeter and 180 mm of bridge infill out of 139 m total. Expect slight drooping on those ceilings, not failure. First layer: 1 continuous island, 372 mm²; islands per layer: median 5, max 7.

4.4 Structural integrity (estimate, not tested)

- Single continuous body; every wall ≥ 1.56 mm (3–4 lines) away from the faces.
- The doubly curved sheet acts as a shell; the weakest direction is between layers (Z), as for any FDM part.
- Thinnest material is at the cube faces and the four vertical edges, where walls are cut at grazing angles – handle by the faces, not the edges.
- No mechanical test or simulation was performed; no load rating is claimed.

4.5 Remaining limitations

- Each 3 × 3 grid is offset a/8 = 3.3 mm from the face centre (inherent to the gyroid; see §2.2).
- Outermost openings keep only a 0.9 mm border to the cube edge.
- Thermal and flow performance were not evaluated for this revision (geometry and printability only).
- Slicing used a generic printer section; re-slice with your own printer profile before printing.

5. Material selection

PLA is selected because the part is a visual and handling prototype: it prints accurately with little warping, bridges and overhangs well with strong cooling (important for the saddle ceilings), needs no enclosure and is widely available.

Assumption	Value	Use / limitation
Density	1.24 g/cm³	Mass estimate
Filament diameter	1.75 mm	Slicer
Nozzle / bed temperature	205–215 °C / 55–60 °C	Typical PLA window; check your spool
Glass transition	≈ 55–60 °C	Do not leave in a hot car or near heat sources
Behaviour	Stiff, brittle	Thin edges can chip if dropped; no impact rating

6. Final slicer configuration (PrusaSlicer, generic printer, PLA, 0.4 mm)

Saved as *print/prusaslicer_PLA_0.4mm_gyroid_3x3.ini*. Settings marked **Geometry** change the physical shape or the size of the openings; the others affect only quality or reliability.

Setting	Recommended	Range	Affects
Nozzle diameter	0.4 mm	0.4–0.6 mm	Quality / time
Layer height (first layer)	0.2 mm (0.2 mm)	0.12–0.24 mm	Surface finish on curved sheets
Line width (first layer)	0.45 mm (0.5 mm)	0.42–0.48 mm	**Geometry**: widths beyond 0.5 mm narrow the openings
Perimeter generator / walls	Arachne, 3 perimeters	3–4	Reliability; walls print as 3–4 lines
Infill	100 % rectilinear	–	Only fills the few wide sheet regions
Top / bottom solid layers	4 / 3	3–5	Saddle ceilings
Nozzle temperature	215 °C first layer, 210 °C	205–220 °C	Reliability
Bed temperature	60 °C	55–65 °C	Adhesion
Speeds	Outer wall 25, walls 40, infill 60, bridges 25, first layer 20 mm/s	Outer 20–35 mm/s	Quality (outer wall speed governs surface)
Cooling	100 % fan from layer 2; bridge fan 100 %	80–100 %	Overhang quality
Retraction	0.8 mm @ 35 mm/s, 0.2 mm Z-hop (direct drive)	Bowden: 4–6 mm	Stringing across openings
Travel	Avoid crossing perimeters ON	–	Keeps strings out of the openings
Adhesion	5 mm outer brim, elephant-foot compensation 0.15 mm	brim 3–8 mm	**Geometry**: compensation keeps bottom openings full size
Supports	**OFF**	(build-plate only if required)	**Geometry**: supports fill openings
XY size compensation	0	0	**Geometry**: any value changes opening size
Orientation	As exported (z = 0 face on the bed)	x = 0 or y = 0 face down is equivalent	Sliced and verified orientation; equivalence by the gyroid's 3-fold symmetry
STL export / resolution	400,000 triangles, decimation < 0.02 % volume change	≥ 300 k triangles	**Geometry**: coarser meshes facet the sheets

0.6 mm nozzle alternative. Because the thinnest printed feature is ≈1.3 mm and the openings are ≈5 mm, a 0.6 mm nozzle (0.3 mm layers, 0.65 mm lines, 2 perimeters) keeps the geometry: PrusaSlicer estimates 1h 10m 0s and 14.95 g, against 1h 57m 47s with 0.4 mm – a meaningful time saving with a coarser surface. 0.4 mm remains the default for surface quality.

7. Support-removal strategy

Supports can be eliminated. The gyroid is a minimal surface: it has no local low points, so every layer grows from the one below except tiny sub-line slivers at the cube edges. The cube is printed as exported (z = 0 face on the bed; the x = 0 and y = 0 faces are equivalent by the gyroid's three-fold x→y→z symmetry, the opposite faces were not sliced). This gives one continuous first layer and no enclosed overhang that needs support.

Slicing case (PrusaSlicer 2.9.6)	Support extruded	Removable (clear to bed)	Resting on model (likely trapped)	Openings after slicing	Time / PLA
Revised – supports OFF (recommended)	0.0 m	0 mm³	0 mm³ (0 pockets)	9 / 9 / 9	1h 57m 47s / 14.78 g
Revised – build-plate only	12.6 m	1,143 mm³	4 mm³ (8 pockets)	9 / 9 / 9	2h 11m 19s / 16.16 g
Revised – everywhere	85.2 m	707 mm³	6,913 mm³ (64 pockets)	9 / 9 / 9	3h 6m 50s / 23.67 g
Original – everywhere	201.1 m	0 mm³	17,233 mm³ (60 pockets)	0 / 0 / 90	10h 36m 47s / 52.98 g

How removability was checked: the support toolpaths were rebuilt from the G-code and each connected support region classified. A region is *clear to bed* when no printed model material lies below it in its column (it stands in a straight vertical void and can be pulled out downwards); otherwise it rests on the lattice's own surfaces inside the cube. With build-plate-only supports 1,143 mm³ is clear to bed and 3.9 mm³ in 8 small regions rests on the model; with 'everywhere' 6,913 mm³ sits in 64 pockets inside the lattice. **Never use 'supports everywhere' on this part.** If a printer struggles with the saddle ceilings, build-plate-only supports are the fallback: almost all of them stand in straight vertical columns open to the bottom face, so geometrically they can be pulled out downwards. This is a geometric classification of the toolpaths, not a tested removal.

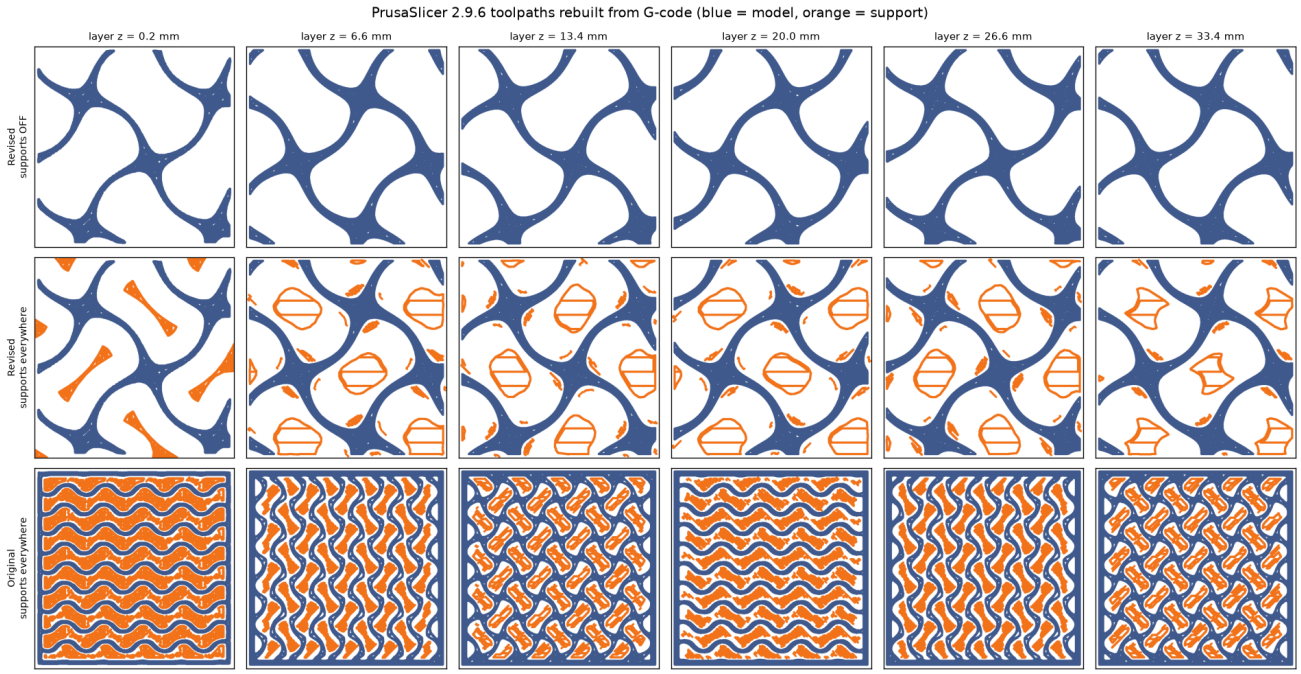


Figure 5 – Toolpaths rebuilt from the PrusaSlicer G-code at six heights (blue = model, orange = support). Top: recommended, no support. Middle/bottom: 'supports everywhere' fills the channels.

8. Validation results

Check	Method	Result	Status
Equation, period, phase, offset	Source inspection (src/gyroid.py, build_3x3.py)	a = 26.667 mm, s = a/8, w = 1.6 mm	Verified
Opening count per axis	Line-of-sight analysis of the voxelised STL (0.1 mm); exact ray–triangle test at sample openings	9 whole, 3 × 3, on X, Y and Z	Verified
Dimensions	STL bounding box	40.000 × 40.000 × 40.000 mm	Verified
Wall thickness	3D medial-axis distance transform; per-layer skeleton widths	p1 1.56 mm	Verified
Mesh integrity	Watertight, manifold, self-intersection, degenerate-face checks	all pass	Verified
Slicing preserves openings	PrusaSlicer 2.9.6 CLI; toolpaths rebuilt from G-code	27 openings, min Ø 4.82 mm	Verified
Support placement / removability	G-code support regions classified	0 support with supports OFF	Verified
Print time / material	PrusaSlicer estimate (generic printer)	1h 57m 47s / 14.78 g	Estimate
Slicer GUI layer preview	–	Not available (command-line slicing only)	Not performed
Physical print / strength	–	–	Not performed

Measurement notes (checked during this work):

- The first voxelisation showed 2 extra 0.2 mm 'pinholes' along X and a 0.2 mm wall minimum. Exact ray–triangle tests showed no line of sight there and that the thin voxels lie outside the mesh. Cause: the analysis slicer interpolated shared edges from opposite ends, giving odd crossing counts on a few scanlines. Fixed (canonical edge interpolation); 9 of 160,000 scanlines still needed neighbour repair. The STL itself was never affected.
- PrusaSlicer first rejected the profile (relative E without a per-layer reset); 'G92 E0' was added to the layer G-code.

9. Final deliverables

File	Format	Content
stl/gyroid_3x3_40mm.stl	Binary STL	Final model, 20.0 MB, SHA-256 f67f57ffb741b91a…
src/build_3x3.py	Python	Editable source: parameters (L, openings per axis, wall), build, validation
src/gyroid.py, mesh_checks.py, openings.py, printability.py	Python	Generator (phase-shift added), mesh checks, opening and layer analysis
src/slice_check.py, render_views.py, make_report.py	Python	Slicer check, views, this report
print/prusaslicer_PLA_0.4mm_gyroid_3x3.ini	PrusaSlicer config	Recommended print settings
results/3x3/validation.json, design_study.json, original_comparison.json	JSON	Measurements
results/3x3/slicer/slicer_report.json (+ *.gcode, not committed)	JSON / G-code	Slicer results; G-code is for the generic printer only
results/3x3/views/*.png	PNG	Front, side, top, isometric, annotated openings, sections, layers
Gyroid_Design_and_3D_Printing_Report.pdf	PDF	This report

10. Final recommendations and workflow

- Open *stl/gyroid_3x3_40mm.stl* in your slicer and keep the exported orientation (z = 0 face on the bed).
- Load *print/prusaslicer_PLA_0.4mm_gyroid_3x3.ini* settings into your own printer profile (keep the print and filament values; keep your printer's start G-code and retraction if it uses a Bowden extruder).
- Make sure supports are **OFF** (not 'everywhere'); XY size compensation 0.
- Slice and inspect the layer preview: 3–4 lines per wall, openings visible in every layer, no support, a single brim around the first layer.
- Print in PLA at 210 °C / 60 °C with 100 % fan from layer 2; expect about 1h 57m 47s and 14.78 g with a 0.4 mm nozzle (generic printer estimate).
- After printing, remove the brim, look through each face: 9 openings must be clear on all six faces.

Pre-print checklist

Done	Check before printing
[ ]	Slicer reports the model as 40.0 × 40.0 × 40.0 mm, manifold, 1 part
[ ]	Supports OFF; 'support everywhere' not selected
[ ]	Perimeter generator Arachne, line width 0.42–0.45 mm, XY compensation 0
[ ]	Elephant-foot compensation ≈ 0.15 mm, 5 mm brim
[ ]	Preview: 9 see-through openings visible on top view, openings open in every layer
[ ]	Cooling 100 % from layer 2; avoid crossing perimeters ON
[ ]	Bed clean and level; first layer continuous
[ ]	After printing: 9 clear openings on each of the six faces

Report generated from the project files on 2026-10-29; STL SHA-256 f67f57ffb741b91a5489f2bd78cb285a91dc4f26cf6f33e192d2a481d2217d7e.